

Sentiment Analysis

From First Principles to Transformers

A comprehensive engineering-driven evaluation of NLP systems comparing classical ML approaches to modern Transformer architectures

 **Mohammad**

Master's of Engineering, University of Ottawa

 December 1, 2025



Project Goal & Problem Statement



Project Goal

Conduct a comprehensive, engineering-driven evaluation of NLP systems, comparing methodologies from classical ML to modern Transformer architectures.

- 📈 Focus on trade-offs between performance and efficiency
- ⚖️ Systematic comparison of diverse architectures
- 🔧 Analysis under real-world engineering constraints



Problem Statement

Despite widespread adoption of models like BERT and GPT, there's a significant gap in understanding:

- Underlying mechanics of NLP systems
- Comparison between classical and modern architectures
- Engineer challenges in optimizing for constrained hardware

"Addresses this with end-to-end empirical analysis of NLP architectures"

Dataset Overview - AG News Corpus

Key Dataset Characteristics



Task

4-class topic classification
World, Sports, Business, Sci/Tech



Size

- ✓ 120,000 training samples
- ✓ 7,600 test samples



Balance

Equal number of documents per class



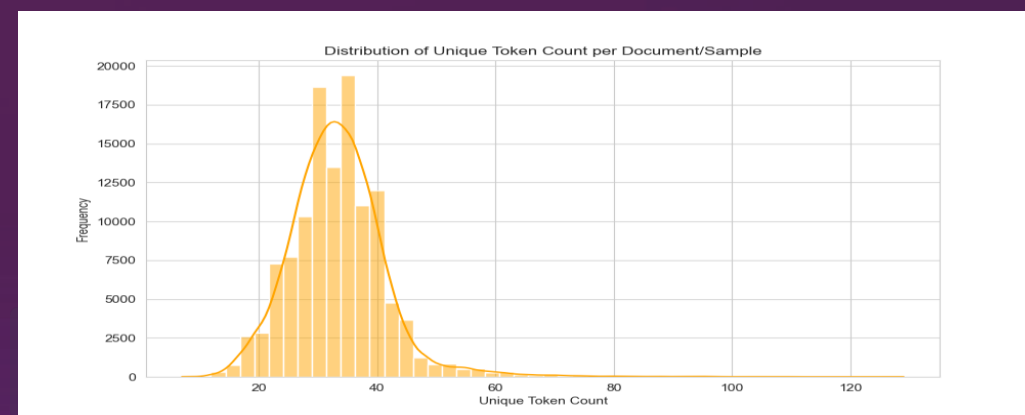
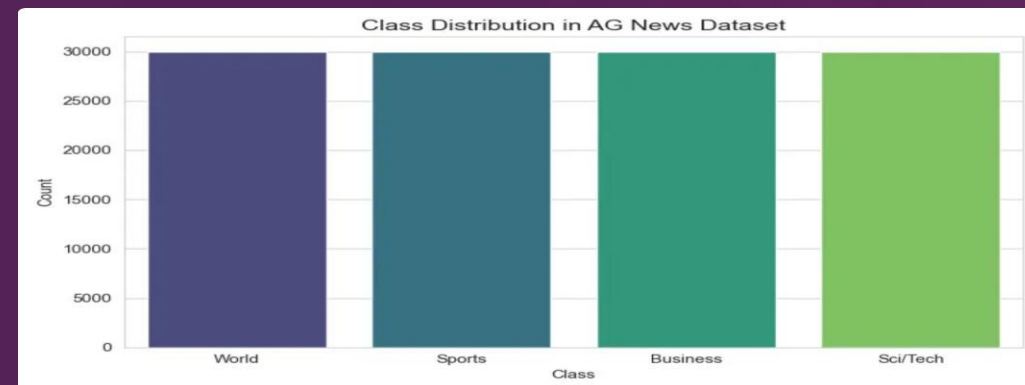
Document Length

Average of 37.85 words



The AG News dataset contains 4 classes: World, Sports, Business, and Science/Technology.

Class Distribution



Classical Machine Learning Approaches

📁 Feature Extraction

📄 Bag-of-Words (BoW)

- ✓ CountVectorizer for 5,000 unigrams
- ✓ Foundation of information retrieval
- ✓ Sparse vector representation

📄 TF-IDF

- ✓ TF-IDF vectorizer with 1-2 grams
- ✓ Max 5,000 features
- ✓ Weighted document vectors

🧠 Classical Models

📈 Logistic Regression

- Linear classifier
- Efficient on linearly separable data

🌲 Random Forest

- Ensemble of decision trees
- Captures non-linear relationships

📊 XGBoost

- Gradient-boosted decision trees
- Enhanced performance over Random Forests

🔄 Workflow

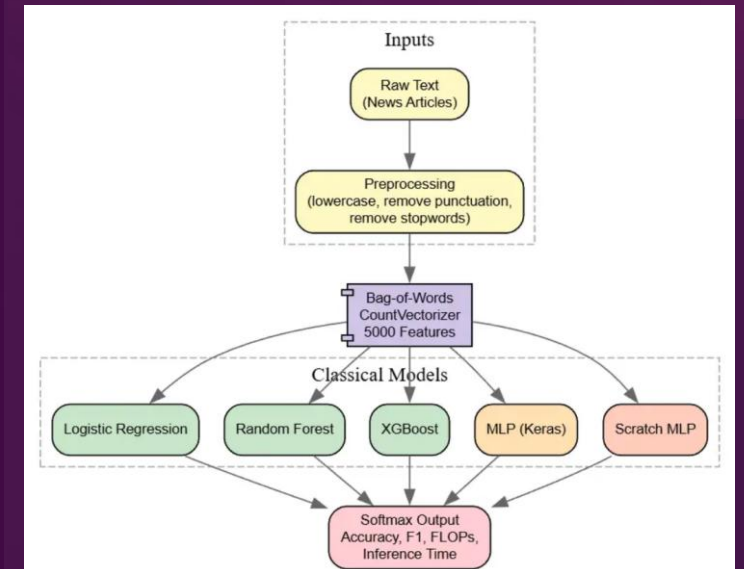


Figure: TF-IDF Vectorizer Workflow for Classical Models

Workflow

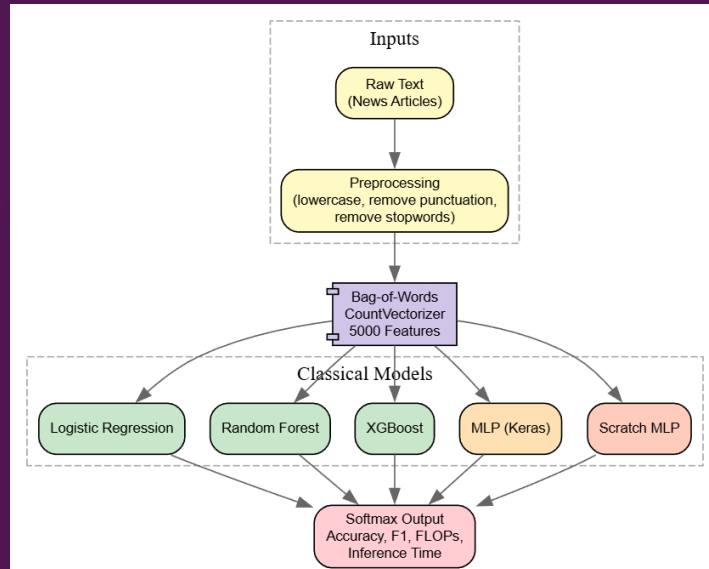


Figure: Bag of Words Vectorizer Workflow for different Models

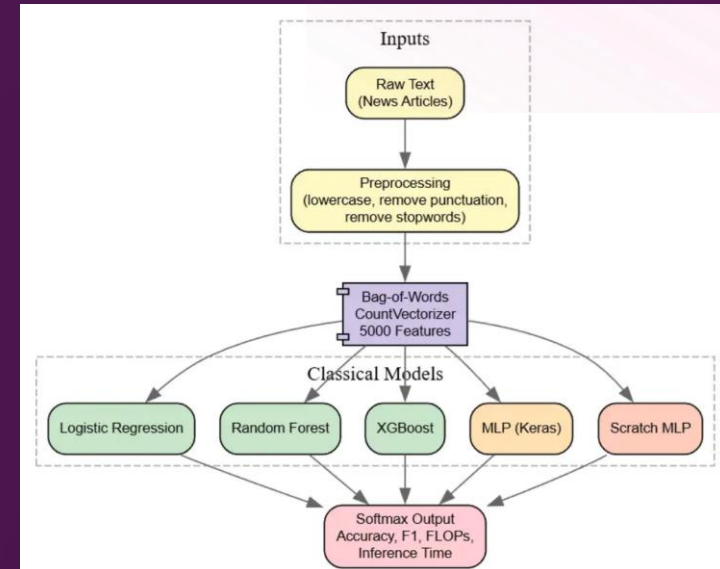


Figure: TF-IDF Vectorizer Workflow for different Models

Embedding-Based Deep Learning Models

📖 Neural Networks as First Principles

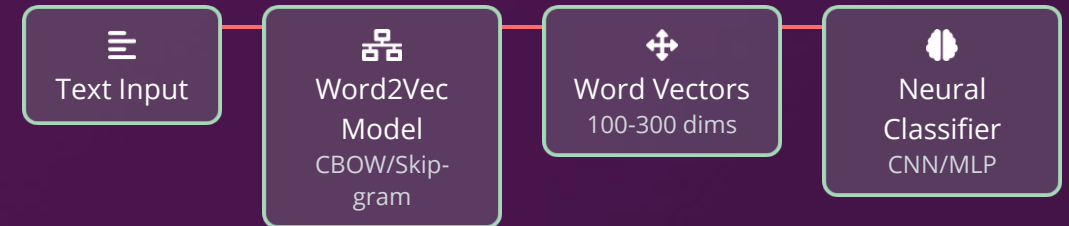
First principles of representation learning through neural embeddings with 100-300 dimensional learnable vector spaces for contextual text.



- ✓ Captures semantic relationships between words
- ✓ Learnable parameters specific to the task

📖 Word2Vec Embeddings as First Principles

Specialized implementation of word vectorization first principles with CBOW or Skip-gram, mapping 100-300 dimensional text concepts into semantic dense vectors.



- ✓ Uses pre-trained embeddings from large corpora
- ✓ Combines well with CNN and MLP classifiers

Transformer Architecture - BERT Fine-tuning



BERT-base-uncased

- ✓ 12-layer, 768-hidden, 12-heads
- ✓ Lowercase tokenizer
- ✓ WordPiece vocabulary



Fine-tuning Configuration

- ⚙️ AdamW optimizer
- 📉 Learning rate: 1e-5
- 🕒 3-5 epochs

BERT Fine-tuning Pipeline



Input Text
Tokenized using
WordPiece



BERT Encoder
Processes tokenized
inputs



[CLS] Token
Extraction of first token



Classification Head
MLP with softmax
output

BERT advances first principles methodologies with multi-headed self-attention, refining attention maps, and pooling embeddings to capture context beyond simple bag-of-words representation

Evaluation Metrics Framework

Predictive Performance

Accuracy

Overall classification correctness measured as the ratio of correctly classified samples to total samples.

F1-Score

Weighted average of precision and recall across the four classes, providing balance between true positives and false positives.

Computational Efficiency

Training Time

Measured in seconds, representing the wall-clock time to train the model on the training dataset.

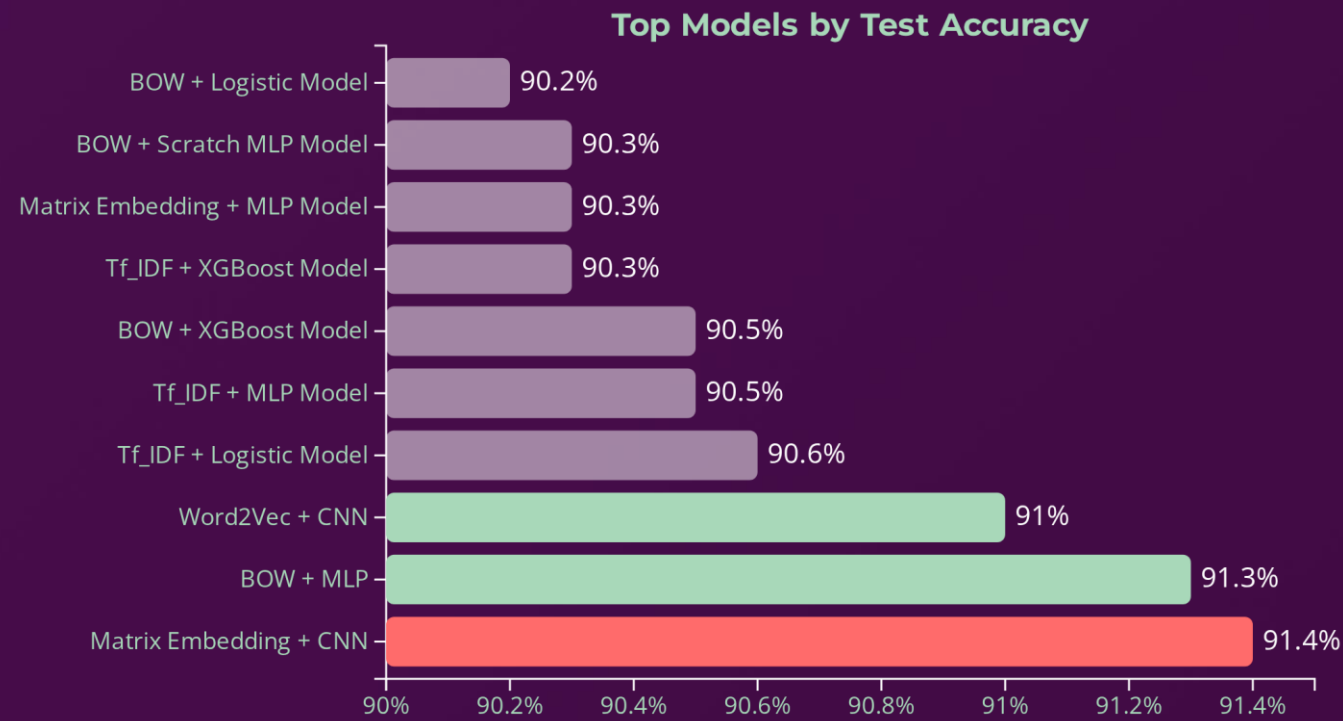
Inference Time

Average time per test sample in seconds, measuring the model's prediction speed.

FLOPs per Sample

Floating Point Operations per second, measuring the computational complexity of the model.

Performance Results Overview



Top Performers

- Matrix Embedding + CNN:** 91.4% accuracy
- BOW + MLP:** 91.3% accuracy
- Word2Vec + CNN:** 91.0% accuracy



Key Insights

- Neural architectures with embeddings outperform classical methods
- MLP models show similar performance to CNNs for this task
- Classical models like Logistic Regression lag behind (90.2% accuracy)

Top Performing Models Analysis

🏆 Top Models by Test Accuracy

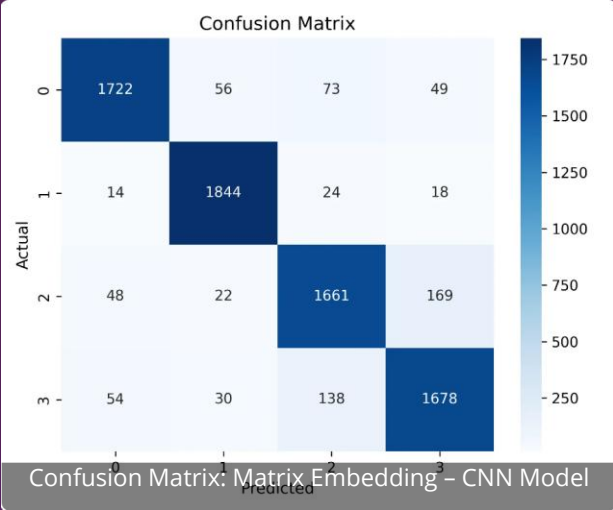
Model	Test Accuracy
Matrix Embedding + CNN	0.914
BOW + MLP	0.913
Word2Vec + CNN	0.910
Tf_IDF + Logistic Model	0.906
BOW + XGBoost Model	0.905

💡 **Key Insight:** Matrix Embedding + CNN achieved the highest accuracy (0.914), demonstrating the effectiveness of combining advanced embedding techniques with neural architectures.

📊 Confusion Matrix Analysis

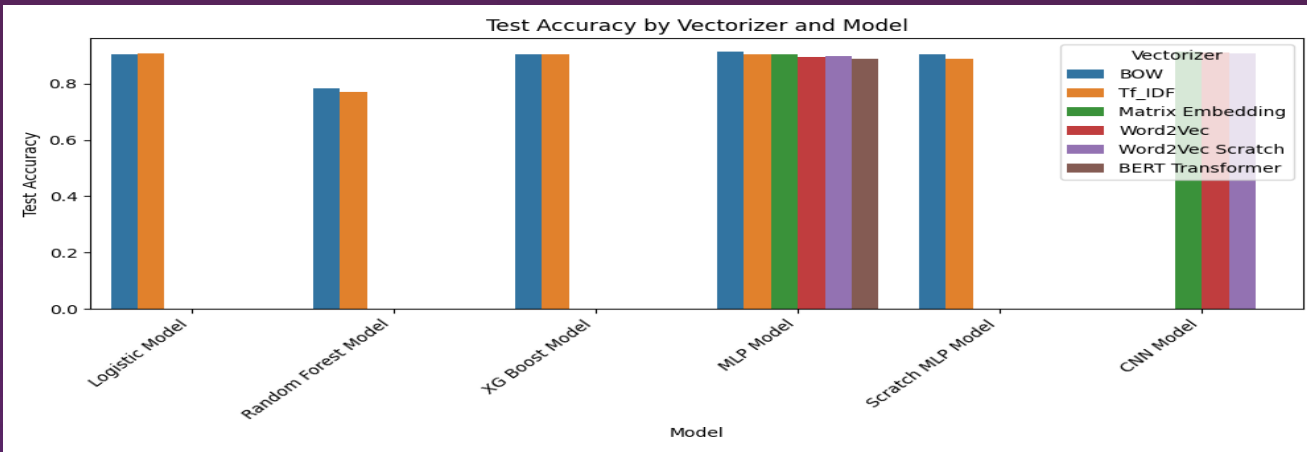
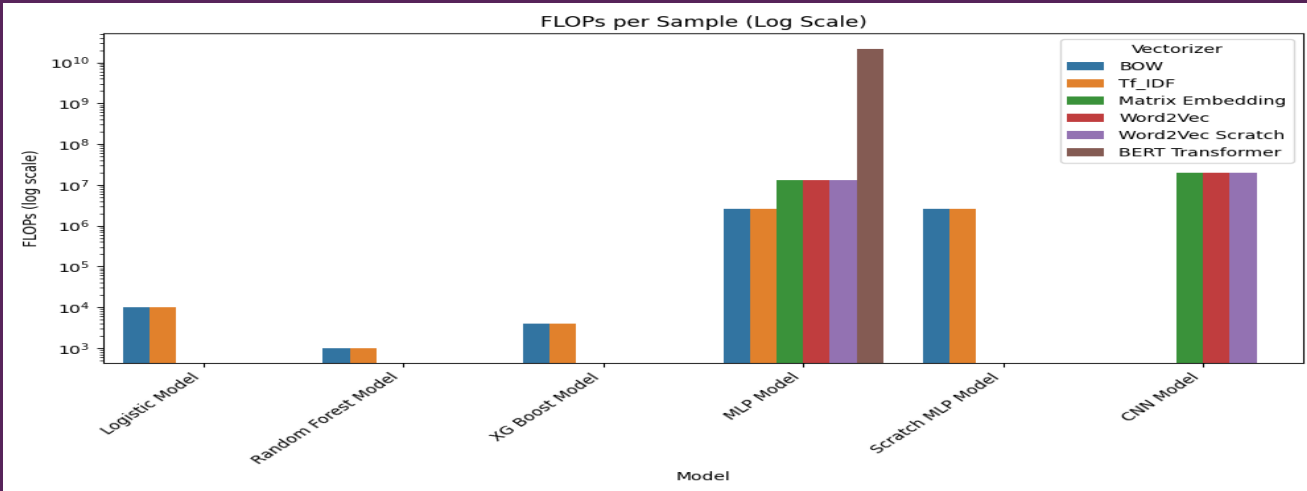
The confusion matrix for the best-performing model (Matrix Embedding + CNN) reveals:

- ✔ Strong classification performance across all news categories
- ✔ Mostly accurate predictions with balanced error distribution
- ✔ Word2Vec model shows slightly better performance (0.893 vs 0.898) on unseen data



Efficiency & Accuracy Trade-offs

FLOPs Per Model



Key Insights

- Efficiency Leaders**
Classical models (BOW/TF-IDF + Logistic) offer excellent efficiency with ~90% accuracy
- BERT Outlier**
BERT has highest computational cost without achieving superior accuracy (0.887)
- Optimal Choice Depends**
Model selection should be guided by deployment constraints and requirements

Efficiency Metrics

- FLOPs per Sample
- Training Time (s)
- Inference Time (s)

Key Findings & Insights



Classical Models as Baselines

Logistic Regression and MLPs on Bag-of-Words/TF-IDF features achieve **~90-91% accuracy** with significantly lower computational costs.

- ✓ Efficient training and inference
- ✓ Strong performance on structured data



Best Accuracy-Cost Balance

Matrix Embedding + CNN achieved **0.914 test accuracy** while maintaining reasonable balance between accuracy, FLOPs, and training time.

- ✓ Suitable for real-time deployment
- ✓ Effective semantic understanding



Model Selection Trade-offs

Optimal model choice dictated by interplay between accuracy, computational efficiency, and deployment constraints.

- ✓ No universal best model
- ✓ Problem-specific optimal choices

💡 Model selection requires balancing accuracy, efficiency, and specific application requirements

Limitations & Future Work

Limitations

➤ Constrained Training Budget

BERT Transformer model underperformed relative to its known capabilities due to limited training resources.

➤ Computational Constraints

Resource limitations prevented exploration of larger model architectures and longer training schedules.

Future Work

Extended Fine-Tuning Schedules

Extend BERT training epochs from 10 epochs to 40-50 to significantly improve accuracy.

Lightweight Transformer Variants

Investigate MobileBERT, TinyBERT, or DistilBERT for resource-constrained environments.

Larger Label Spaces

Evaluate transformers on tasks with hundreds or thousands of output classes to highlight advantages.

Thank You

Questions?



Mohammad

Master's of Engineering, University
of Ottawa



fmoaha077@uottawa.ca



December 2, 2025



Project Repository

GitHub



https://github.com/mohammad1774/applied_ml_project



Feedback and discussions welcome